

Project Proposal

CMPT310 D200 Summer 2026

Group 8

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Project Idea

We will build an AI-image detector using a convolutional neural network that takes an image input and returns a real number between 0 and 1 representing the likelihood of the input image being AI generated or altered. The detector helps users determine whether an image was created or changed by artificial intelligence, improving trust in digital media.

First, we will train our model by defining the features of AI and non-AI images, then we will tweak our model until it achieves acceptable accuracy. Finally, we will create a UI server to host an interface for the program that can be accessed on a web browser.

Tools and Resources

Python Libraries

- PyTorch: For training the model
- OpenCV: Access to computer vision functions
- PIL: For image processing
- NumPy: To work with matrices
- Pandas: To handle large datasets
- ONNX: To export the model
- Flask: To serve the user interface

Data

- CIFAKE: contains 60,000 real images and 60,000 mirrored synthetic images.

- Access at <https://github.com/jordan-bird/CIFAKE-Real-and-AI-Generated-Synthetic-Images>
- Real and Fake (AI-Generated) Art Images Dataset: 10,821 images of authentic art and 10,821 AI-generated images.
 - Access at <https://www.kaggle.com/datasets/doctorstrange420/real-and-fake-ai-generated-art-images-dataset>

Project Plan / Timeline

Milestone 1: July 1st

Model can predict if an image is AI-made or altered to an acceptable accuracy using Boolean classification.

Milestone 2: July 29th

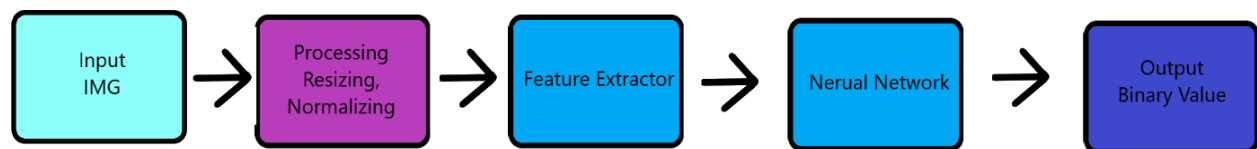
Model now uses regression to determine a likelihood percentage of an image being AI created or altered and can be accessed via a web interface.

Minimal Viable System

The minimal viable system is a convolutional neural network that can make a binary classification as to whether an image is AI altered or created to an acceptable accuracy.

We will first prioritize training the model to be as accurate as possible. Since we live in an age of disinformation, having an easy way to check with a high level of confidence that an image is produced by a computer or not could be very useful to many people. Our system could be used in the future to prevent fraud, deception, or detect deep fakes.

Minimum Viable System



Desired System

